

Igor Varejão

igorvarejao.tech | igor.varejao.dev@gmail.com | [Linkedin](#) | [Github](#)

EXPERIENCE

Data Engineer

Olho Do Dono

October 2024 – May 2025

Vitória, ES, Brazil

- Created and maintained CI/CD pipelines, supporting DevOps practices for automated testing and deployment
- Utilized Azure, Docker, and PostgreSQL for infrastructure and data processing support
- Scaled cattle weight prediction systems and optimized computer vision workflows for production environments

Data Engineer Intern

Olho Do Dono

July 2022 – September 2024

Vitória, ES, Brazil

- Developed cattle weight prediction systems based on videos using Python, ensuring performance and scalability
- Performed image annotation, segmentation, tagging, and data ingestion in CVAT with rigorous data quality/sanity checks; while overseeing a team executing the same annotation and segmentation workflows.
- Conducted extensive Exploratory Data Analysis (EDA) using Jupyter Notebook, Briefer, and Plotly to assess and optimize the performance of AI systems
- Contributed in the development and maintenance of computer vision and AI solutions, coordinating multiple servers through ZMQ for efficient communication.

Artificial Intelligence Undergraduate Researcher

Federal University of Espírito Santo

August 2020 – July 2024

Vitória, ES, Brazil

- Researched and developed deep neural networks for defect pattern recognition in submersible centrifugal pumping systems to the DRPDBCS project in partnership with Petrobras
- Performed exploratory data analysis (EDA) using Jupyter Notebook and Plotly to visualize patterns and derive actionable insights.
- Applied data preprocessing and manipulation techniques to vibration signals across time, frequency, and time–frequency domains using Python (NumPy, Pandas)
- Developed a public framework to centralize public datasets using Python for deep learning research in defect pattern recognition at github vibdata repository.
- Trained deep neural networks using frameworks pytorch, pytorch-lightning and scikit-learning
- Conducted comprehensive statistical tests (e.g., t-test, z-test, Wilcoxon, Welch's) to rigorously evaluate the significance and validity of research findings.
- Authored and published a paper in one of the top journals in the field, *Journal of Sound and Vibration*, titled *The similarity bias problem: What it is and how it impacts vibration based intelligent fault diagnosis* as a result of this research.

Artificial Intelligence Volunteer Researcher

Federal University of Espírito Santo

August 2024 – December 2024

Vitória, ES, Brazil

- Developed deep neural network models using transfer learning for fault diagnosis in rotary machinery bearings, in collaboration with the Mechanical Department at the University of Ottawa, Canada.
- Built a benchmarking framework to evaluate a pre-trained model on centralized public bearing datasets, comparing its performance against state-of-the-art pretraining approaches.
- Published research paper *Towards a Universal Vibration Analysis Dataset A Framework for Transfer Learning in Predictive Maintenance and Structural Health Monitoring*

EDUCATION

Federal University of Minas Gerais

Master of Science in Computer Science, Artificial Intelligence

Belo Horizonte, MG, Brazil

August 2025 – Present

- Current research focus: AI Music Generation & MIR

Federal University of Espirito Santo

Bachelor in Computer Science

Vitória, ES, Brazil

March 2019 – November 2024

TEACHING EXPERIENCE

Teaching Assistant – Supervised Machine Learning

March 2026 – June 2026

Federal University of Minas Gerais

Belo Horizonte, MG, Brazil

- Served as teaching assistant for a Supervised Machine Learning course in the Data Science Specialization program covering KNN, Linear and Logistic Regression, SVM and Decision Tree
- Conducted tutorial sessions and office hours to help students understand complex deep learning concepts and implementation techniques
- Instructed students on implementing foundational machine learning models from scratch using NumPy, alongside practical applications using scikit-learn.

Teaching Assistant – Neural Networks and Deep Learning

August 2025 – December 2025

Federal University of Minas Gerais

Belo Horizonte, MG, Brazil

- Served as teaching assistant for a Deep Learning course in the Data Science Specialization program covering Neural Networks Training, Convolutional Networks, and Recurrent Neural Networks
- Conducted tutorial sessions and office hours to help students understand complex deep learning concepts and implementation techniques
- Provided guidance on PyTorch-based implementations and best practices for deep learning model development

Teaching Assistant – Competitive Programming (INF16169)

March 2023 – July 2023

Federal University of Espírito Santo

Vitória, ES, Brazil

- Served as teaching assistant for the Competitive Programming elective course in the Computer Science Bachelor program
- Mentored students in algorithmic problem-solving techniques, data structures, and optimization strategies using C++ and Python
- Evaluated student submissions and provided detailed feedback on algorithm efficiency and code quality

PERSONAL PROJECTS

News WebScraper (Github) | *Python, Selenium*

May 2022 – July 2023

- Developed a web scraping pipeline to collect a textual corpus from a newspaper website for the study of neology in digital culture
- Implemented Selenium to automate page navigation and extraction of web content
- Utilized BeautifulSoup to parse, clean, and structure the textual data for corpus construction
- Optimized the scraping and data processing workflow to ensure efficiency and timely completion of the project